EB2 – supported J1939 Acceleration Modes (TSC1-TE)

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Message		Identifier (hex)	Bytes	Scale
Speed Request	TSC1-TE	0C000003	Bytes 2-3	0,125rpm/bit
Throttle Request	TSC1-TE	0C000003	Byte 4	1%/bit,offset

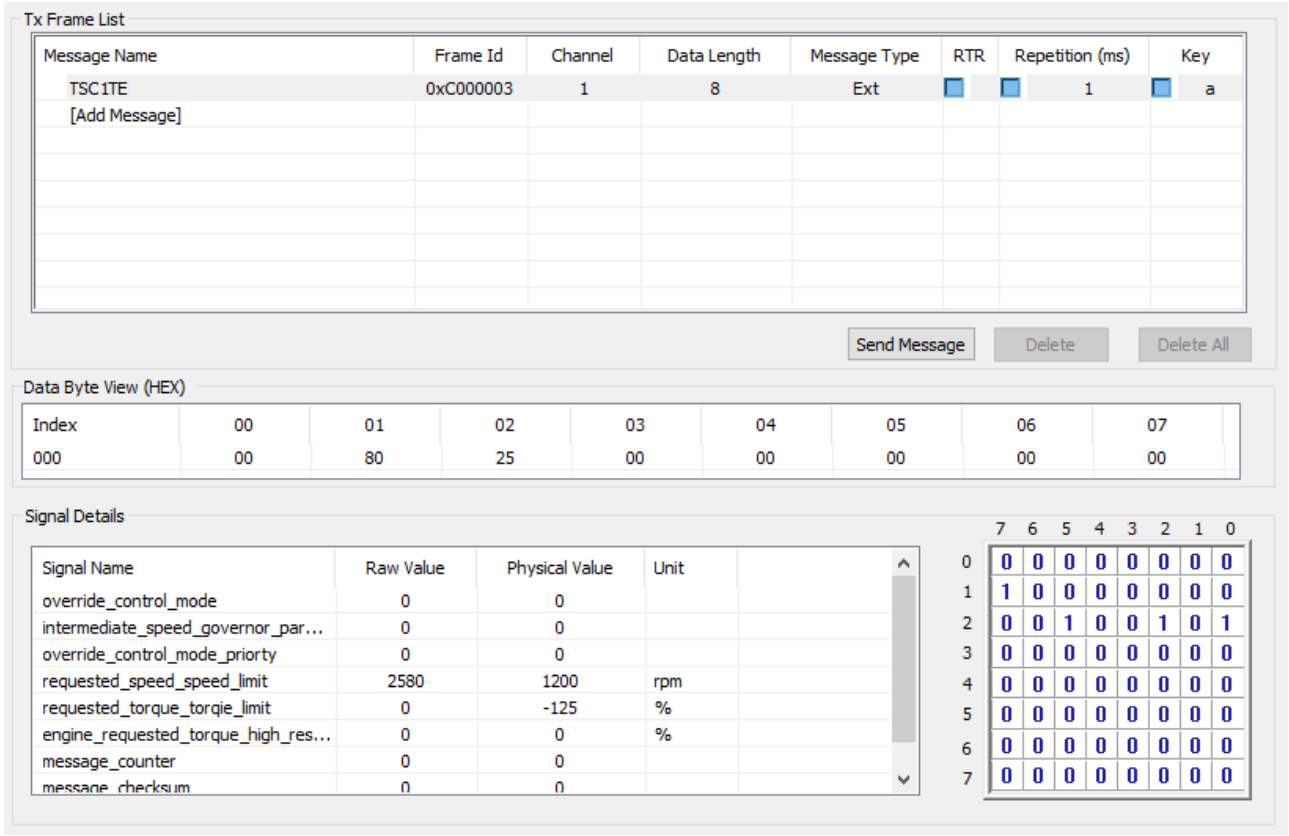
EB2 – supported J1939 Acceleration Modes (TSC1-TE) Requested Speed

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Parameter	Control Bits	Requested Speed/ Speed Limit		Requested Torque / Torque Limit	Not Used	Extension Byte	Not Used	Message Counter / Checksum
SAE J1939/71 Reference		898						
Scale		0.125 rpm/bit						
Offset		0 rpm						
Data Range		0 to +8031.875 rpm						
Operating Range		0 to +8031.875 rpm						
Remarks		Demanded speed. Out of operating range values are not considered FFh in case Not Used						

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Requested Speed - *EXAMPLE*

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Example while using Busmaster Software:



Tx Frame List

Message Name	Frame Id	Channel	Data Length	Message Type	RTR	Repetition (ms)	Key
TSC1TE	0xC000003	1	8	Ext	☐	☐ 1	☐ a
[Add Message]							

Send Message | Delete | Delete All

Data Byte View (HEX)

Index	00	01	02	03	04	05	06	07
000	00	80	25	00	00	00	00	00

Signal Details

Signal Name	Raw Value	Physical Value	Unit
override_control_mode	0	0	
intermediate_speed_governor_par...	0	0	
override_control_mode_priorty	0	0	
requested_speed_speed_limit	2580	1200	rpm
requested_torque_torqie_limit	0	-125	%
engine_requested_torque_high_res...	0	0	%
message_counter	0	0	
message_checksum	0	0	

	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0
2	0	0	1	0	0	1	0	1
3	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0

EB2 – supported J1939 Acceleration Modes (TSC1-TE)

Throttle Request

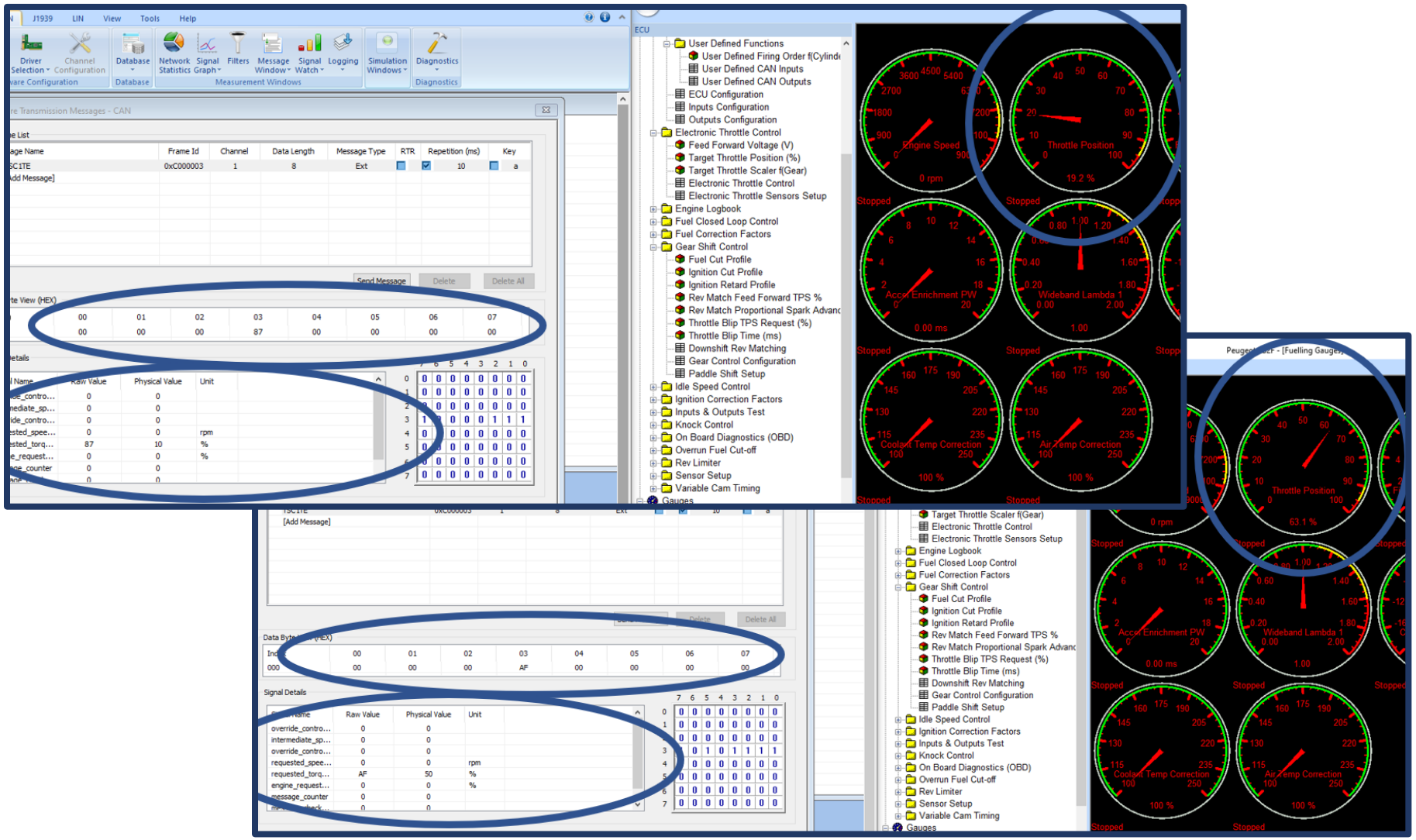
	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Parameter	Control Bits	Requested Speed/ Speed Limit		Requested Torque / Torque Limit	Not Used	Extension Byte	Not Used	Message Counter / Checksum
SAE J1939/71 Reference				518				
Scale				1%/bit gain				
Offset				-125%				
Data Range				-125 to +125%				
Operating Range				0 to +125%				
Remarks				Demanded positive torque, as a percent of reference engine torque (SPN 544) Out of operating range values are not considered FFh in case Not Used				

Requested Torque is converted to throttle demand by the software and calibration.

The scaling is as per J1939 SPN518 Data Length: 1 byte Resolution: 1 %/bit, -125 % offset

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Throttle Request - *EXAMPLE*



%- TSC1 Message	Real %-Throttle Position on Engine
10	19,2
15	25,1
20	30,6
25	35,7
30	40,8
35	46,7
40	52,2
45	57,3
50	63,1
55	67,5
60	71,8
65	76,1
70	81,2
75	86,7
80	91,4
85	96,9
90	99,6